PROOFS

A proof is a logical argument that shows a statement is true for all values in the domain.

COMMON PROOFS

METHOD	HOW IT WORKS
Direct Proof	Start with the given hypotheses and use algebra, definitions, and previously proven results to show the conclusion is true.
Proof by Contrapositive	To prove "if p, then q," prove the contrapositive: "if not q, then not p." That is, prove $\neg q \Rightarrow \neg p$.
Proof by Contradiction	Assume the statement is false. Show this assumption leads to a contradiction (a false statement). Therefore, the original statement must be true.
Proof by Cases	When the domain can be divided into two or more cases (often involving absolute value or piecewise definitions), prove the statement is true in each case.
Mathematical Induction	Used for statements about integers $n \geq n_0$. 1. **Base Step:** Prove true for $n = n_0$. 2. **Inductive Step:** Assume true for $n = k \geq n_0$. Prove true for $n = k + 1$. Then it is true for all $n \geq n_0$.

EXAMPLE: PROOF BY CONTRADICTION

Prove: $\sqrt{2}$ is irrational.

Proof:

Assume $\sqrt{2}$ is rational. Then $\sqrt{2} = \frac{a}{b}$ where a, b are integers with no common factor. Then $2 = \frac{a^2}{b^2} \Rightarrow a^2 = 2b^2$.

So a^2 is even, and thus a is even. Let $a = 2k$.

Then $2b^2 = a^2 = (2k)^2 = 4k^2 \Rightarrow b^2 = 2k^2$.

So b^2 is even, and thus b is even.

Thus a and b are both even, contradicting that they have no common factor. Therefore, $\sqrt{2}$ is irrational.

KEY LOGIC

$p \Rightarrow q \equiv \neg q \Rightarrow \neg p$

$\neg(p \wedge q) \equiv \neg p \vee \neg q$

$\neg(p \vee q) \equiv \neg p \wedge \neg q$

$\neg(\neg p) \equiv p$

$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$

$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$

TIPS FOR WRITING PROOFS

- Start by clearly stating what you are proving.
- Use given information, definitions, and previously proven theorems.
- Each step must follow logically from a previous step.
- Conclude by restating what has been proven.

LIMITS

The limit of $f(x)$ as x approaches a is the value L that $f(x)$ approaches as x gets arbitrarily close to a (not necessarily equal to a).

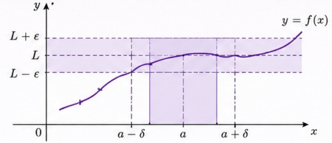
DEFINITION

$$\lim_{x \to a} f(x) = L$$

means that for every $\varepsilon > 0$, there exists $\delta > 0$ such that if $0 < |x - a| < \delta$, then

$$|f(x) - L| < \varepsilon.$$

GRAPHICAL IDEA



LIMIT LAWS

1	$\lim_{x \to a} (c) = c$	Constant Law
2	$\lim_{x \to a} x = a$	Identity Law
3	$\lim_{x \to a} [c\, f(x)] = c \lim_{x \to a} f(x)$	Constant Multiple Law
4	$\lim_{x \to a} [f(x) \pm g(x)] = \lim_{x \to a} f(x) \pm \lim_{x \to a} g(x)$	Sum / Difference Law
5	$\lim_{x \to a} [f(x)\, g(x)] = \lim_{x \to a} f(x) \ \lim_{x \to a} g(x)$	Product Law
6	$\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{\lim_{x \to a} f(x)}{\lim_{x \to a} g(x)}$ if $\lim_{x \to a} g(x) \neq 0$	Quotient Law
7	$\lim_{x \to a} [f(x)]^n = \left[\lim_{x \to a} f(x)\right]^n$	Power Law
8	$\lim_{x \to a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \to a} f(x)}$ (for odd n; for even n, require limit > 0)	Root Law

IMPORTANT LIMITS

$$\lim_{x \to a} c = c$$

$$\lim_{x \to a} x = a$$

$$\lim_{x \to 0} \frac{\sin x}{x} = 1$$

$$\lim_{x \to 0} \frac{1 - \cos x}{x} = 0$$

$$\lim_{x \to 0} \frac{\tan x}{x} = 1$$

$$\lim_{x \to 0} \frac{a^x - 1}{x} = \ln a$$

$(a > 0)$

TECHNIQUES FOR EVALUATING LIMITS

- **Direct Substitution:** Plug in $x = a$. If defined, the limit equals the value.
- **Factor and Cancel:** Simplify expressions to remove common factors.
- **Rationalize:** Use conjugates to remove radicals.
- **Special Limits:** Use important limits listed above.
- **L'Hôpital's Rule:** For $\frac{0}{0}$ or $\frac{\infty}{\infty}$ forms (in calculus).

FORMS TO CHECK

- $0/0$
- ∞/∞
- $\infty - \infty$
- $0 \cdot \infty$
- 0^0, 1^∞, ∞^0